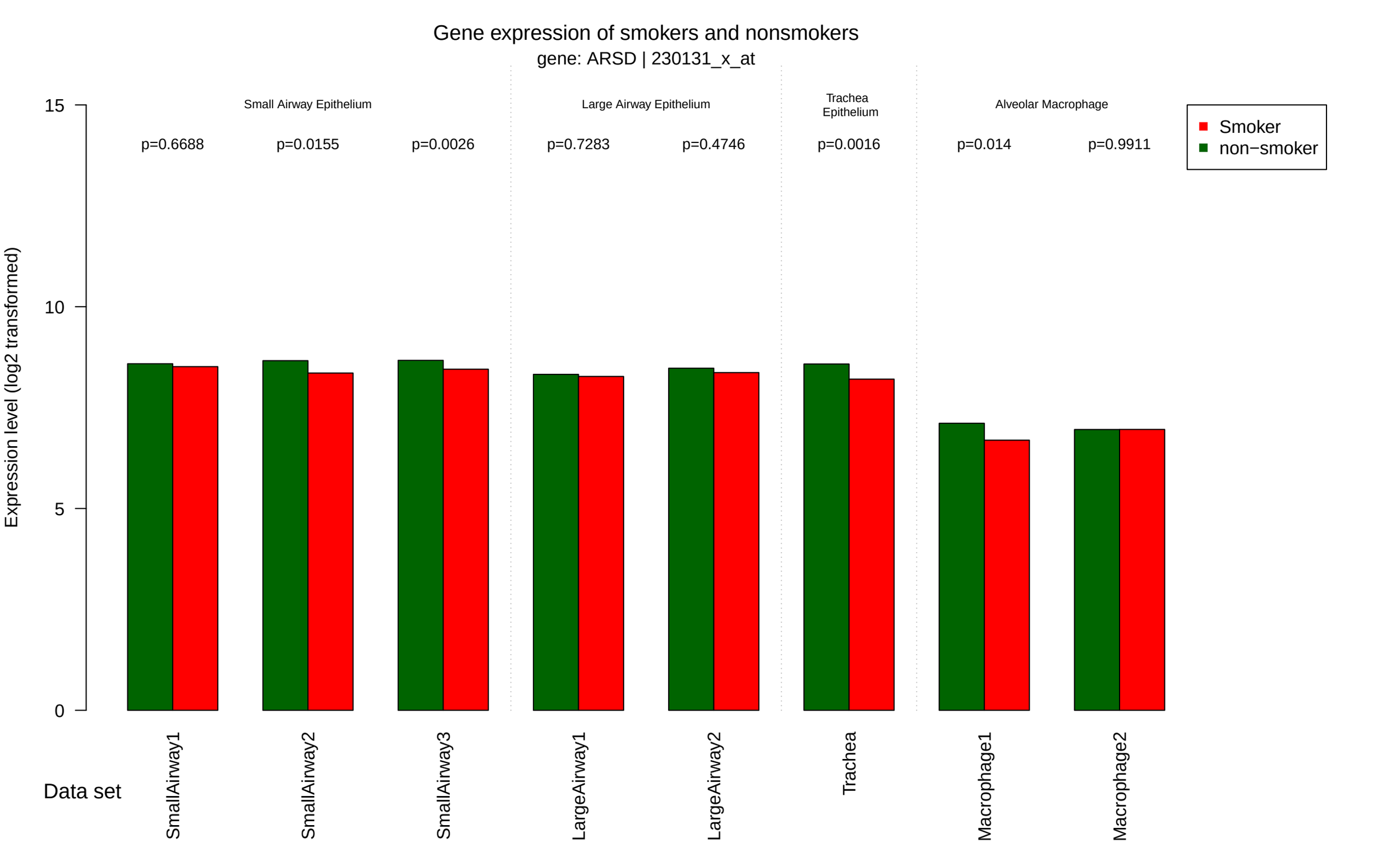
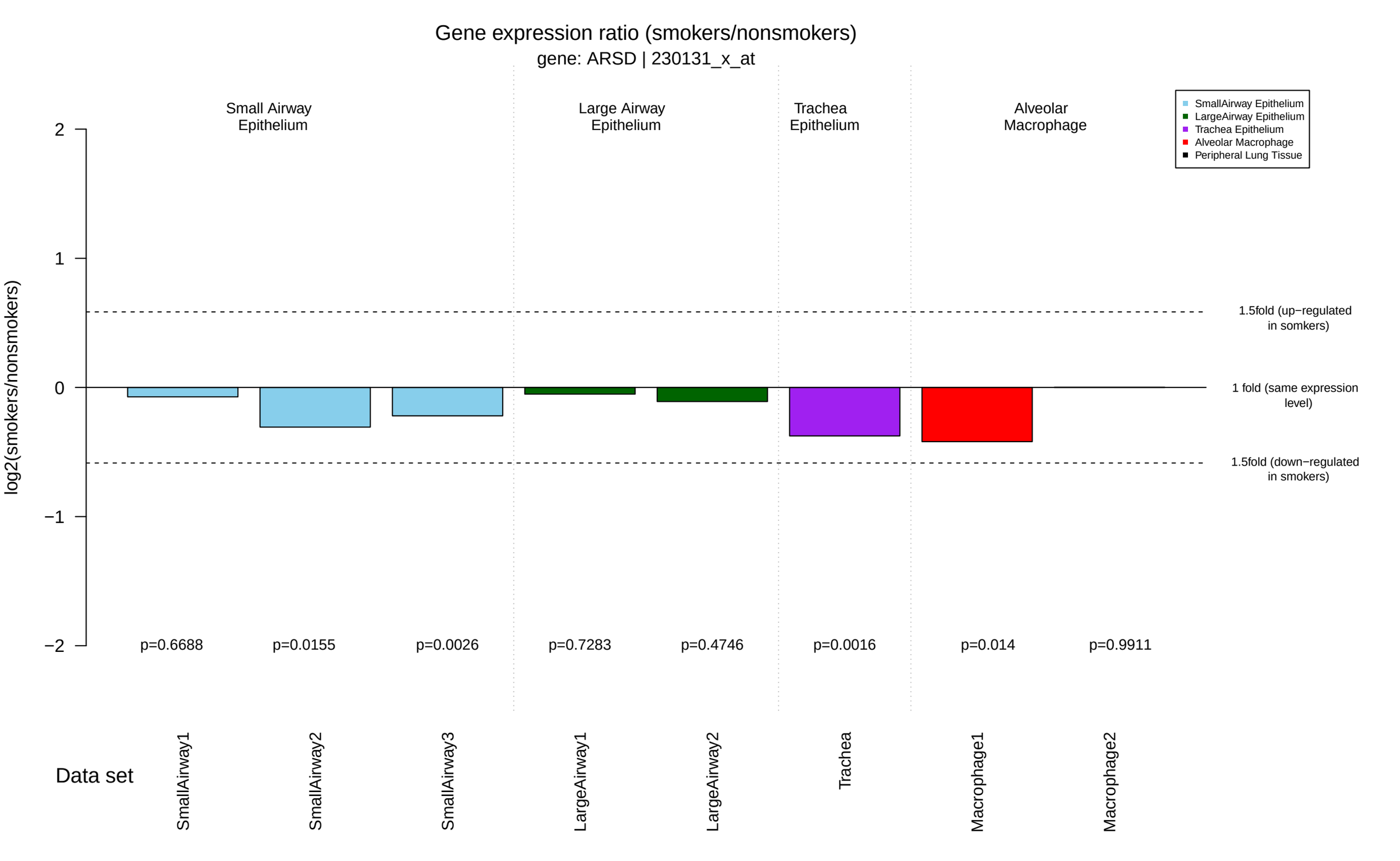
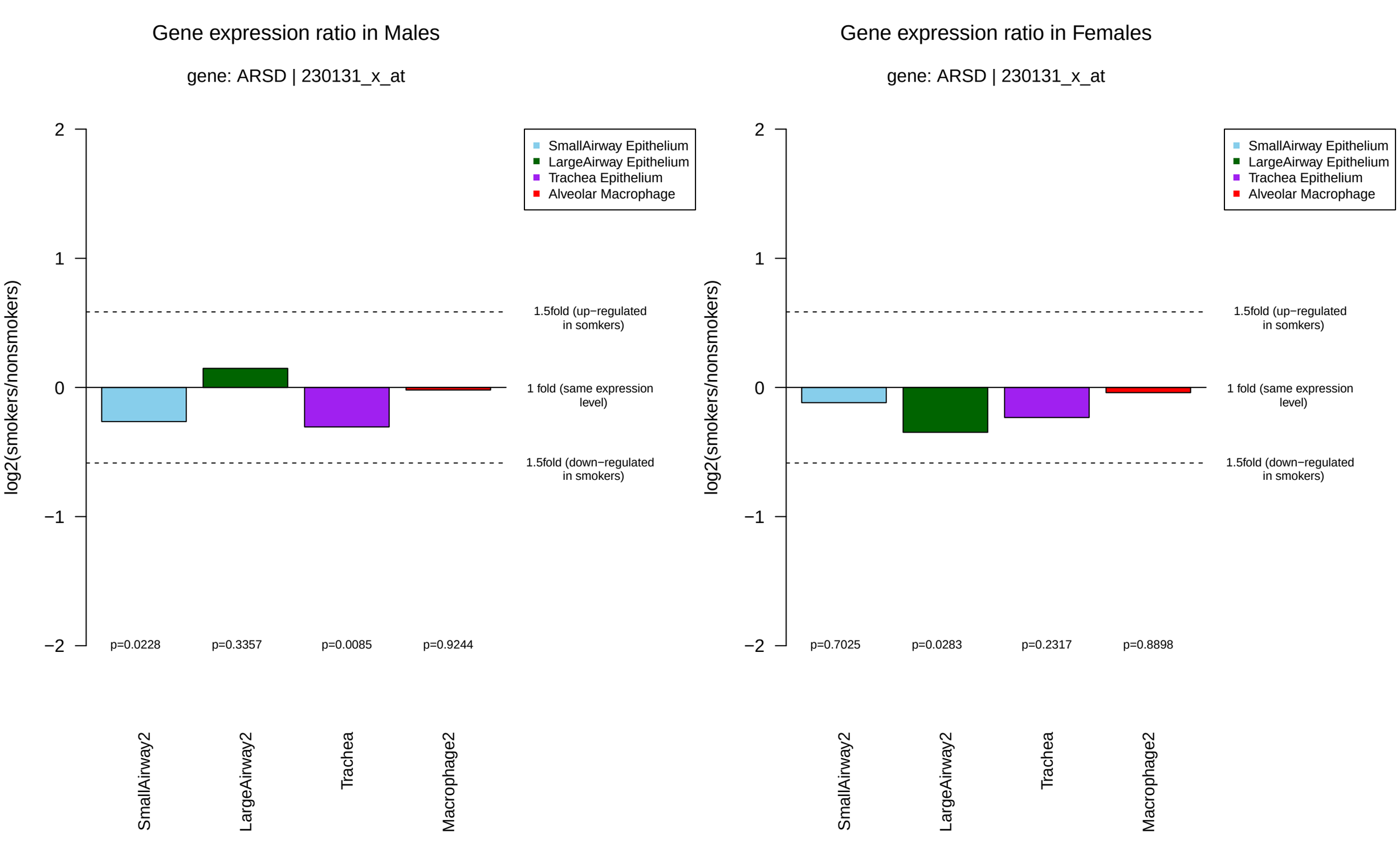
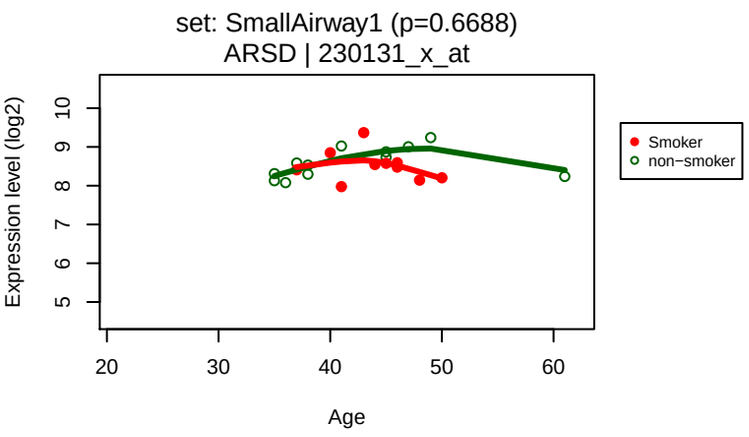
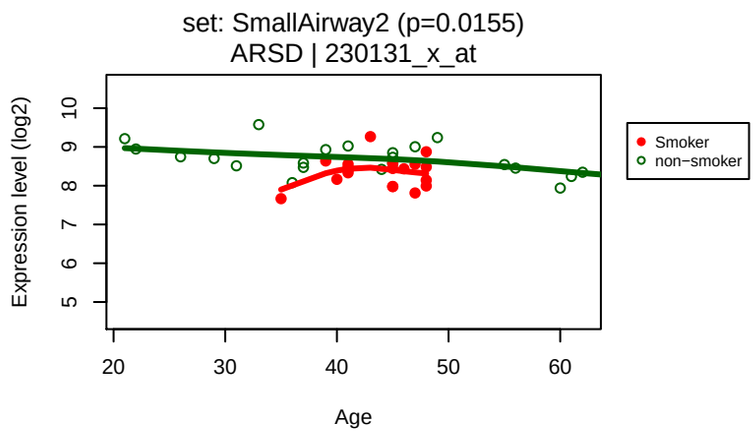
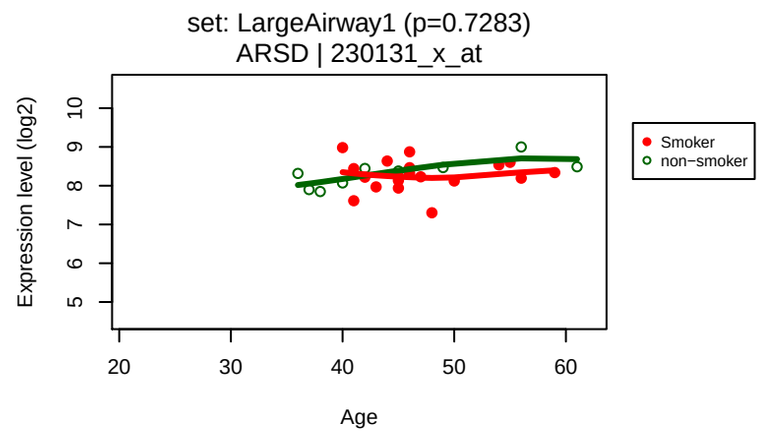

Gene expression of smokers and nonsmokers
gene: ARSD | 230131_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6688
p=0.0155
p=0.0026
p=0.7283
p=0.4746
p=0.0016
p=0.014
p=0.9911
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

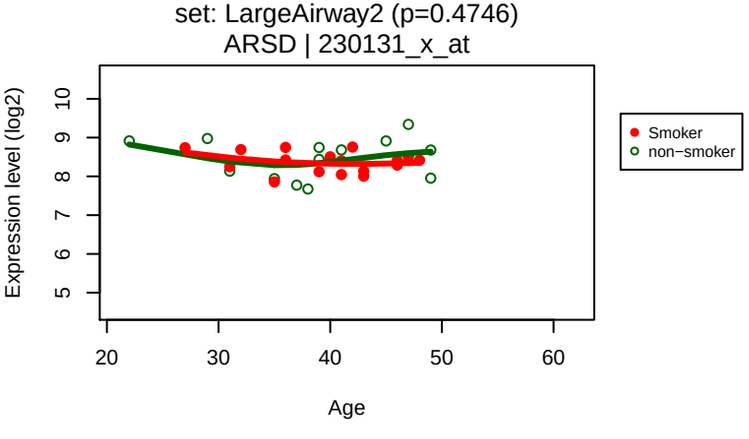

Gene expression ratio (smokers/nonsmokers)
gene: ARSD | 230131_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6688
p=0.0155
p=0.0026
p=0.7283
p=0.4746
p=0.0016
p=0.014
p=0.9911
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

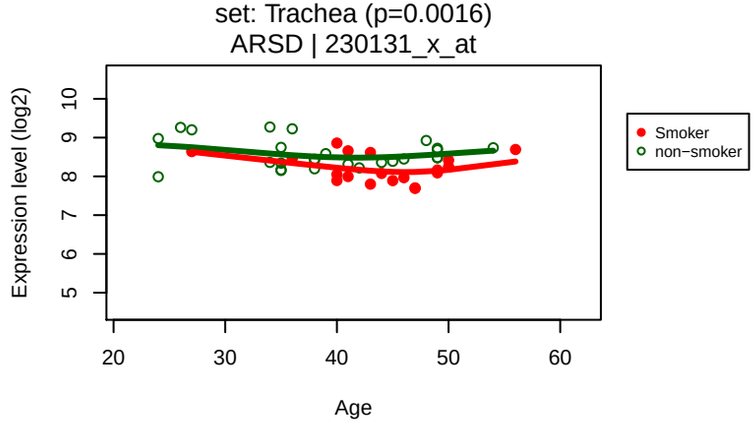

Gene expression ratio in Males
gene: ARSD | 230131_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0228
p=0.3357
p=0.0085
p=0.9244
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARSD | 230131_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7025
p=0.0283
p=0.2317
p=0.8898
SmallAirway2
LargeAirway2
Trachea
Macrophage2

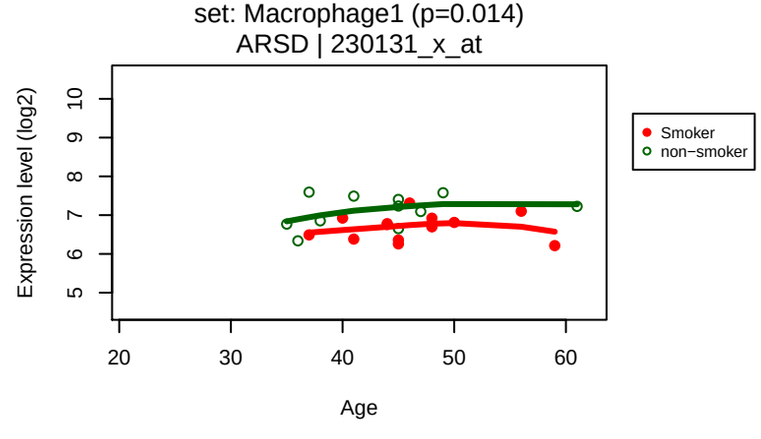

set: SmallAirway1 (p=0.6688)
ARSD | 230131_x_at
Expression level (log2)
Age
Smoker
non−smoker

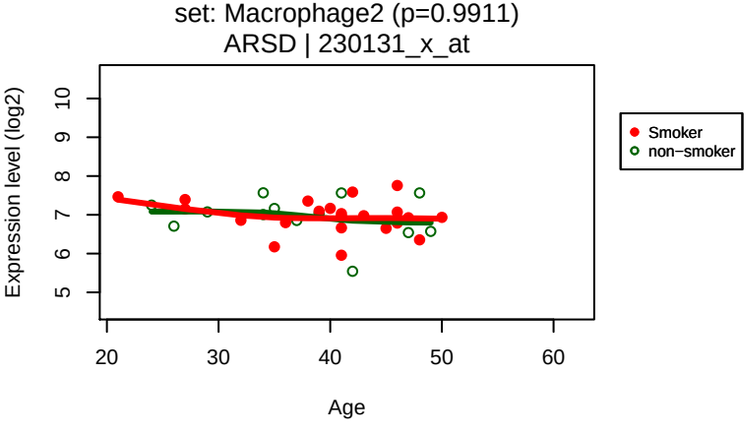

set: SmallAirway2 (p=0.0155)
ARSD | 230131_x_at
Expression level (log2)
Age
Smoker
non−smoker

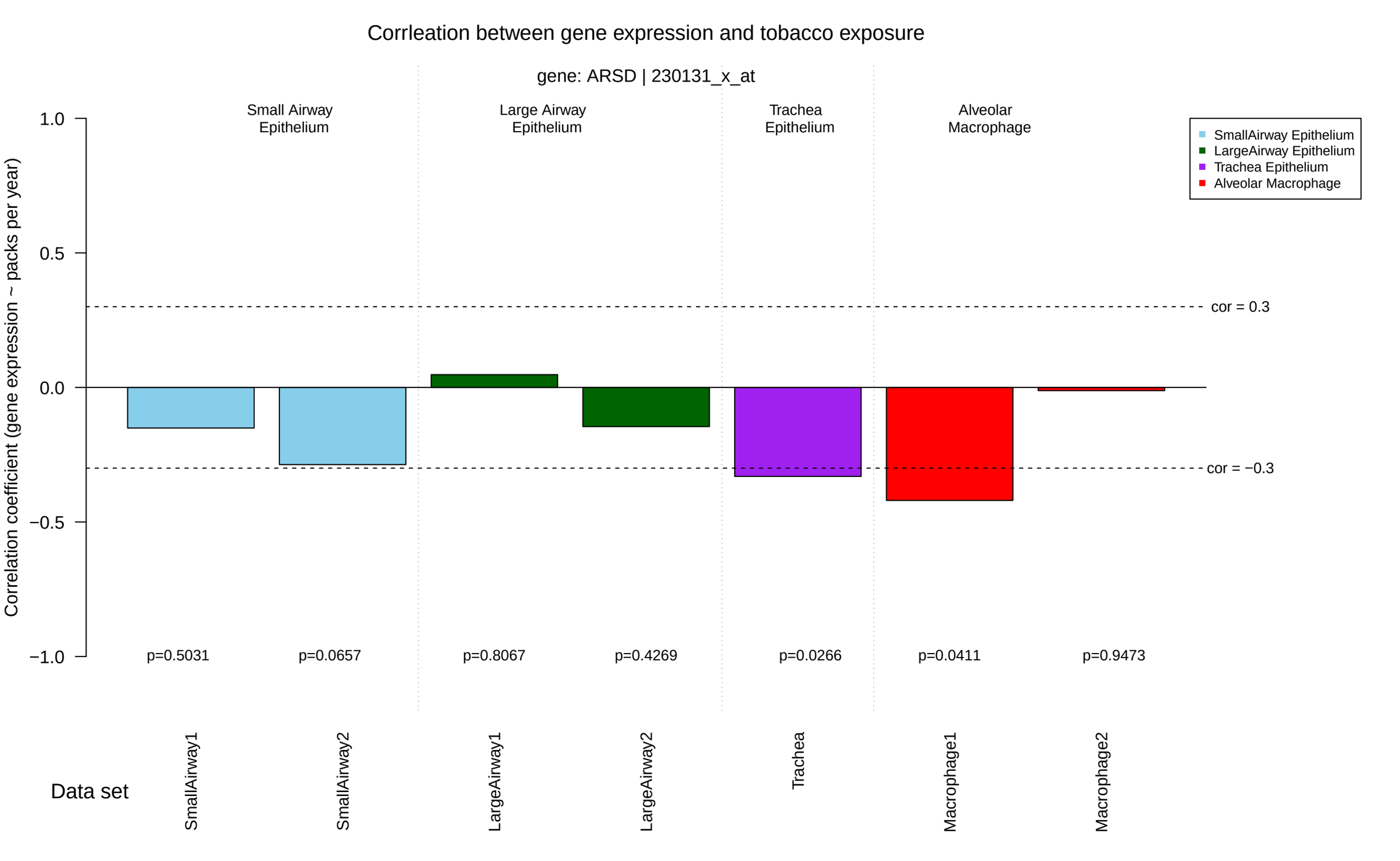

set: LargeAirway1 (p=0.7283)
ARSD | 230131_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4746)
ARSD | 230131_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0016)
ARSD | 230131_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.014)
ARSD | 230131_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9911)
ARSD | 230131_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARSD | 230131_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5031
p=0.0657
p=0.8067
p=0.4269
p=0.0266
p=0.0411
p=0.9473
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2